**GigaWhat Energy**

GigaWhat Gas
Controlled procedure

# PR-GAS-066 Odorant Level Checks

| Field | Value |
|---|---|
| Document ID | PR-GAS-066 |
| Version | 2 |
| Effective from | 2024-08-05 |
| Review due | 2027-08-05 |
| Owner | Head of Gas Network Operations |
| Approved by | Gas Safety Manager |
| Applies to | GigaWhat Pressure Reduction Station (GPR-GWT): pressure reduction station PRS-GWT-01 |

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

## 1 Purpose

Natural gas has no smell of its own. Odorant is added so that people can smell a gas escape well before it becomes dangerous. This procedure sets out how odorant levels are checked at GPR-GWT, the acceptable ranges, and what to do when levels are low.

## 2 Scope

This procedure applies to the supplementary odorant injection unit that forms part of PRS-GWT-01, its odorant storage tank, and the odorant sampling point on the station outlet. Gas arrives at GPR-GWT already odorised upstream; the injection unit tops up the odorant when the incoming level is low.

## 3 References

- PR-GAS-040 Working in Hazardous Areas (DSEAR Zones)
- PR-GAS-021 Permit-to-Work for Gas Operations
- PR-GAS-045 Leakage Survey of Gas Mains
- PR-CORP-001 Reporting Incidents and Near Misses
- Gas Safety (Management) Regulations 1996 and HSE guidance L80
- Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) and HSE guidance L138

## 4 Definitions

Odorant: The mercaptan-based liquid blend used by GigaWhat Gas to give natural gas its smell.

Odorant concentration: The amount of odorant in the gas, in milligrams per cubic metre (mg/m3), measured with a portable odorant analyser at the outlet sampling point.

Odour intensity check: A test by two trained assessors using an odorometer to confirm that gas diluted with air has a distinct smell.

## 5 Responsibilities

Competent Person (Pressure Systems): Carries out the checks, records the results, adjusts the injection rate within the limits in 6.1 and arranges refills.

Gas Network Controller: Is told before any odorant refill or sampling that may release odour, and records low readings.

Authorised Person (Gas): Approves injection rate changes outside the normal range and any work on the injection unit under PR-GAS-021.

Gas Safety Manager: Decides on the wider response to a critical low odorant level.

## 6 Procedure

### 6.1 Acceptable ranges

| Check | Frequency | Acceptable range | Low action level | Critical level |
|---|---|---|---|---|
| Odorant concentration at outlet | Monthly | 5.0 to 10.0 mg/m3 (target 7.0) | Below 5.0 mg/m3 | Below 3.5 mg/m3 |
| Odour intensity | Monthly | Distinct odour at 1% gas-in-air (20% LEL) | Faint odour at 1% gas-in-air | No odour at 1% gas-in-air |
| Odorant tank level | Weekly | 30% to 85% | Below 30% | Below 15% |

> NOTE: Concentrations above 12.0 mg/m3 are also reported to the Gas Network Controller, as over-odorisation leads to false reports of gas escapes.

### 6.2 Weekly tank check

6.2.1 Tell the Gas Network Controller you are on site, and follow PR-GAS-040 inside the station compound.

6.2.2 Read the odorant tank sight gauge and record the level as a percentage.

6.2.3 Read the injection unit display and record the injection rate and any alarms.

6.2.4 Check the tank, pipework and bund for leaks, stains or strong odour.

6.2.5 If the level is below 30%, order a refill for delivery within 5 working days.

### 6.3 Monthly concentration and odour intensity checks

6.3.1 Check that the odorant analyser has been calibrated within the last 12 months.

6.3.2 Purge the sampling line for 30 seconds, then take three readings 2 minutes apart at the outlet sampling point.

6.3.3 Record the average of the three readings as the odorant concentration.

6.3.4 Carry out the odour intensity check with two trained assessors, using the odorometer set to 1% gas-in-air.

6.3.5 Record both results and compare them with the ranges in 6.1.

### 6.4 Actions when odorant is low

> CAUTION: Under-odorised gas may not be smelt at a leak. Treat a critical result as a public safety matter.

6.4.1 Below 5.0 mg/m3 or faint odour: tell the Gas Network Controller, increase the injection rate by no more than 20%, and re-sample within 24 hours.

6.4.2 If the re-sample is still below 5.0 mg/m3, ask the Authorised Person (Gas) to approve a further increase and check the injection unit for faults.

6.4.3 Below 3.5 mg/m3 or no odour: tell the Gas Network Controller and the Gas Safety Manager immediately, and sample daily until the level is back within range.

6.4.4 The Gas Safety Manager decides on further action, which may include extra leakage surveys under PR-GAS-045 and informing the upstream gas transporter.

6.4.5 Tank level below 15%: arrange an emergency refill the same day.

### 6.5 Handling odorant and refills

> WARNING: Odorant is highly flammable and has a very strong smell. A small spill can cause many public reports of gas escapes.

6.5.1 Tell the Gas Network Controller before every refill, so that public reports of a gas smell can be expected.

6.5.2 Wear chemical-resistant gloves, goggles and overalls when handling odorant.

6.5.3 Keep a spill kit with absorbent and odour neutraliser at the tank during refills.

6.5.4 Contain any spill with absorbent, apply neutraliser, and seal waste in the containers provided.

### 6.6 Emergency

6.6.1 Call 999 for fire or injury.

6.6.2 Report a large odorant spill to the Gas Network Controller, or call the National Gas Emergency Service on 0800 111 999 if the Controller cannot be reached.

6.6.3 Report every spill and every critical low result under PR-CORP-001.

## 7 Records

The odorant record holds weekly tank levels and injection rates, monthly concentrations and odour intensity results, refill dates and volumes, and actions taken. It is kept in the GPR-GWT asset file for 6 years.

## 8 Revision history

| Version | Date | Change |
|---|---|---|
| 1 | 2020-08-03 | First issue. |
| 2 | 2024-08-05 | Added monthly odour intensity check and critical level of 3.5 mg/m3. Tank refill trigger raised from 20% to 30%. |